Science, Mathematics and Technology

SSILP Math

Beginner

Polar coordinates

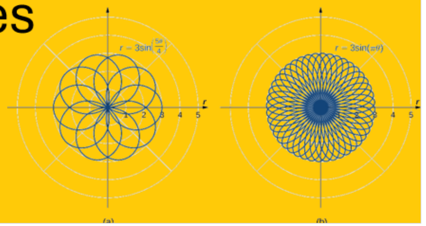
Learning objectives

- Understand the concept of using polar coordinates, including their advantages for describing curves with polar affinity.
- Compare and contrast Cartesian and polar coordinate systems, recognizing the representations of points in each system.
- Learn the conventions and definitions of polar coordinates, including how to specify points using distance and angle, and how to handle negative angles and radii.
- Develop skills in converting between polar and rectangular coordinates, ensuring correct quadrant considerations and understanding the relationship between the two systems.
- Gain proficiency in converting equations between polar and rectangular forms and sketching common polar curves.

Session outline

4

1. Polar coordinates



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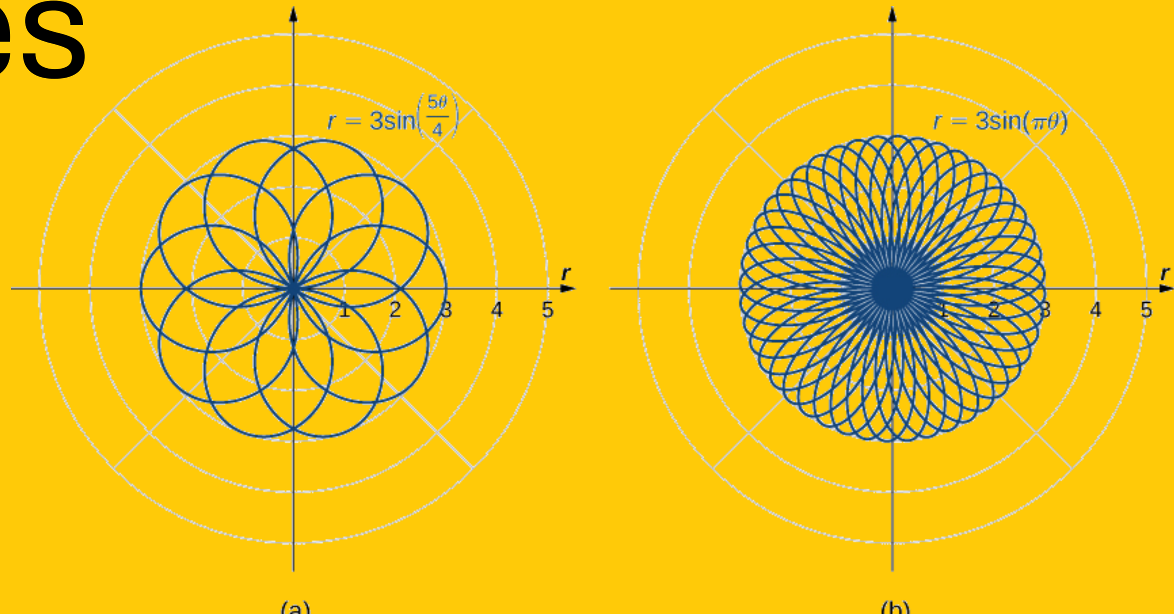
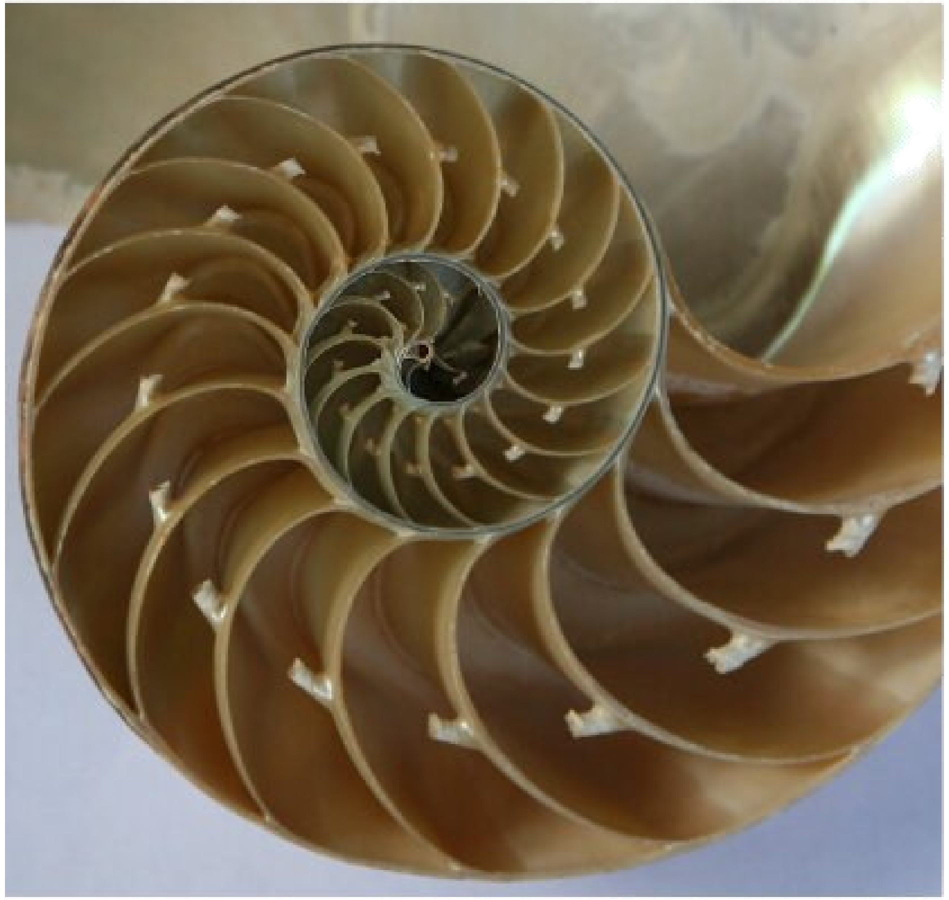
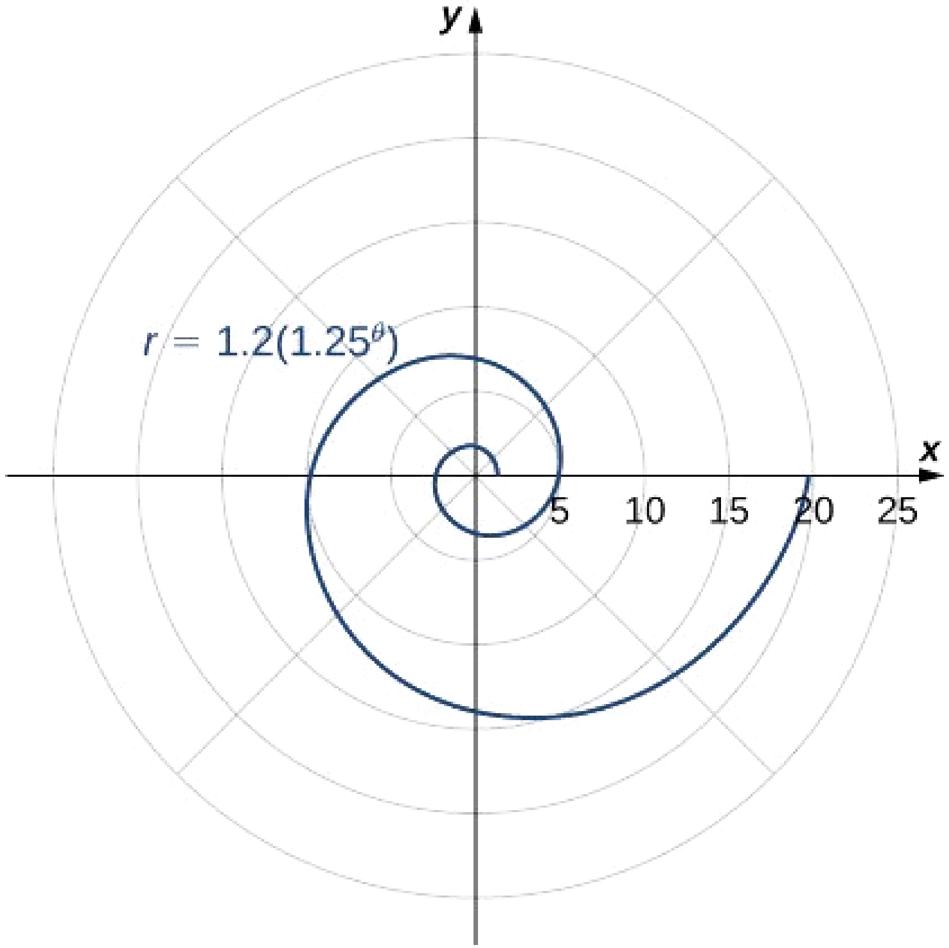
2. Graphs of polar equations

1. Polar coordinates



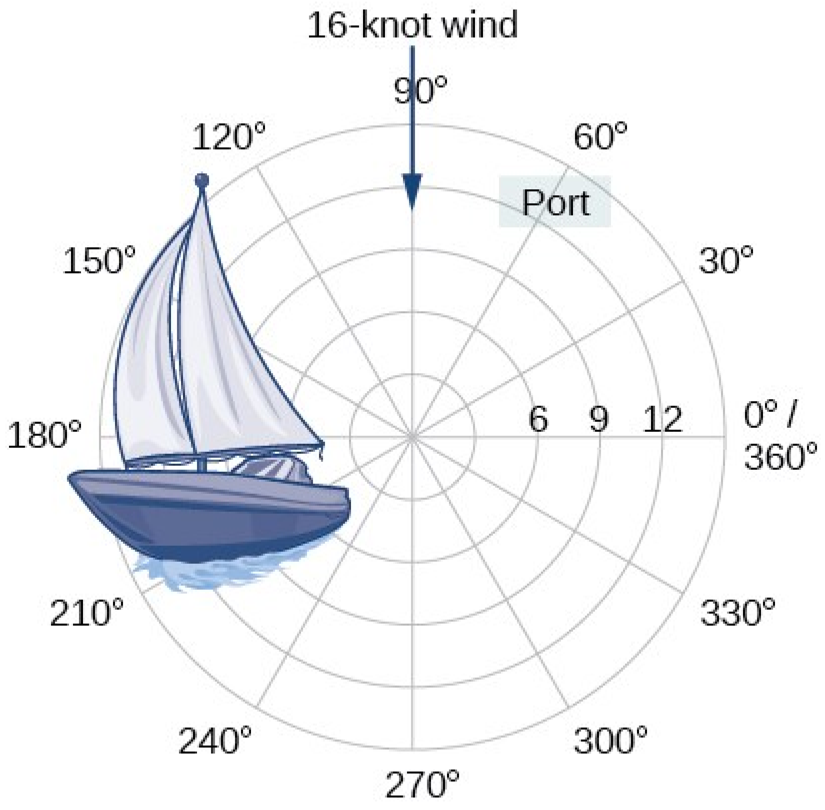
Motivation





Certain **curves** are **more naturally described** in a coordinate system that is based on circles rather than rectangles...

Motivation

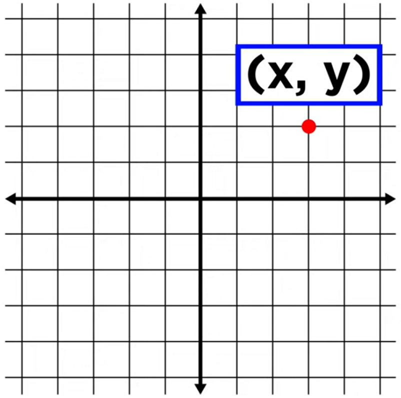


Over 12 kilometers from port, a sailboat encounters rough weather and is blown off course by a 16-knot wind.

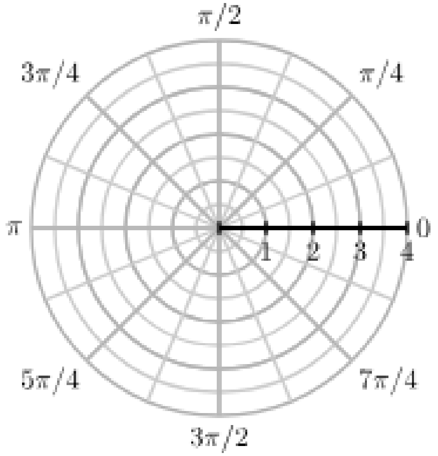
How can the sailor **indicate his location** to the Coast Guard?

Certain **problems** are also **more naturally described** in a coordinate system that is based on circles and directions rather than rectangles…

Cartesian coordinates



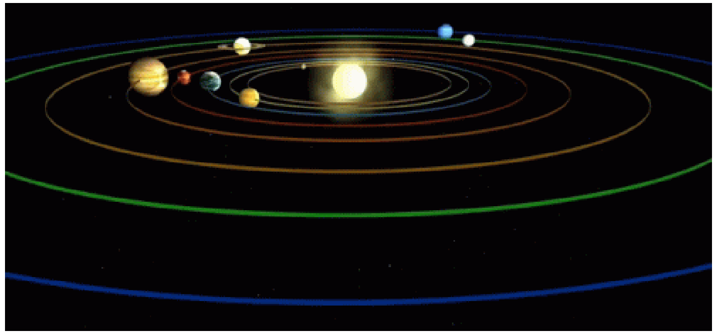
Polar coordinates

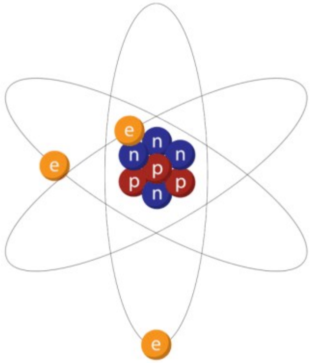


- Suited for graphs and regions that are naturally considered over a rectangular grid.

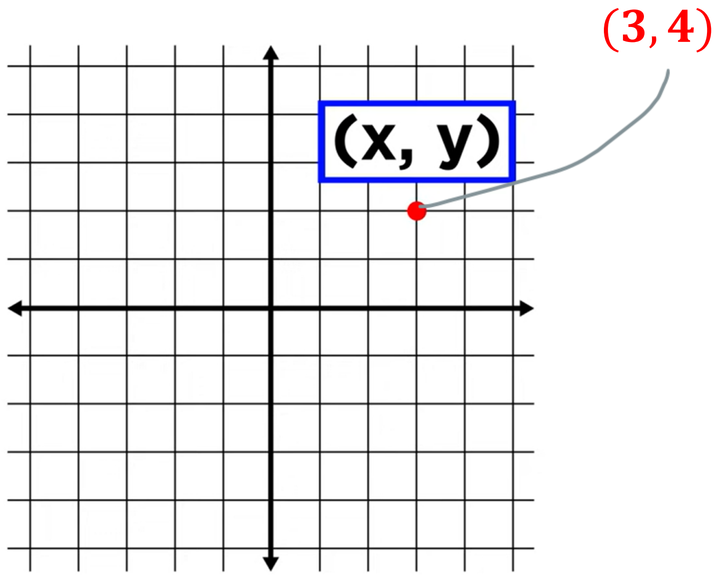
- Domains less suited to rectangular coordinates, such as circles.
- Curves that have an **affinity to a centre**.

Why do we even need different coordinate systems?



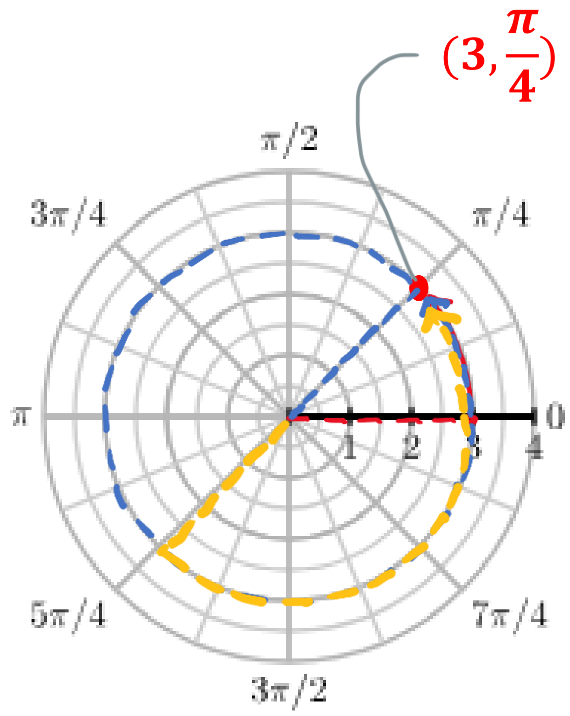


Cartesian coordinates



- Expressing pts on a **rectangular grid**
- Positions indicated by ordered pair corresponding to an x-coordinate and a y-coordinate
- Pts can have only one set of rect coordinates

Polar coordinates



(r, θ)

r = radius

(distance from origin)

θ = angle

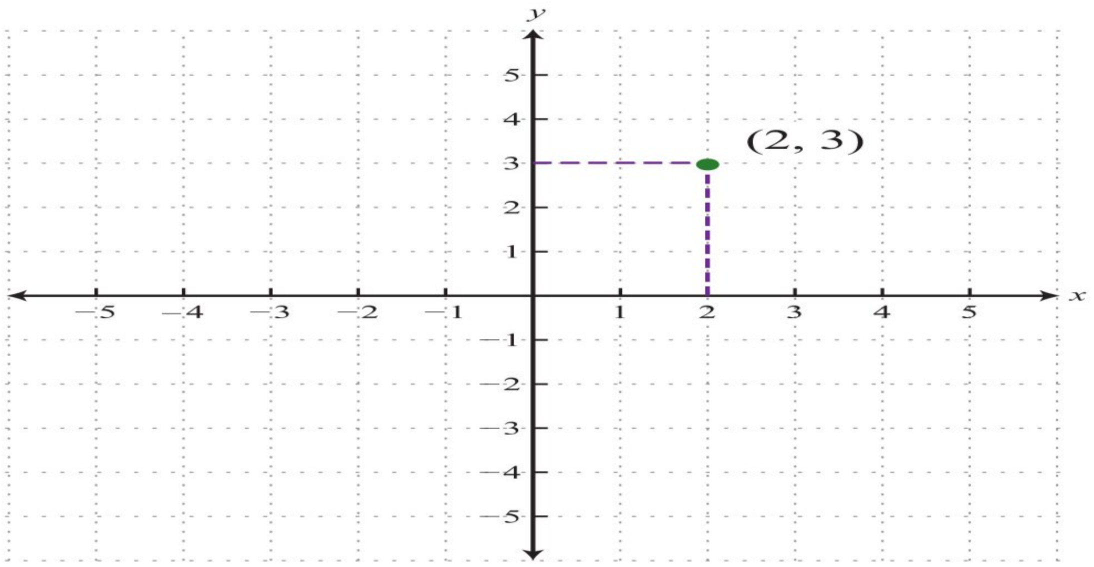
(from polar axis)

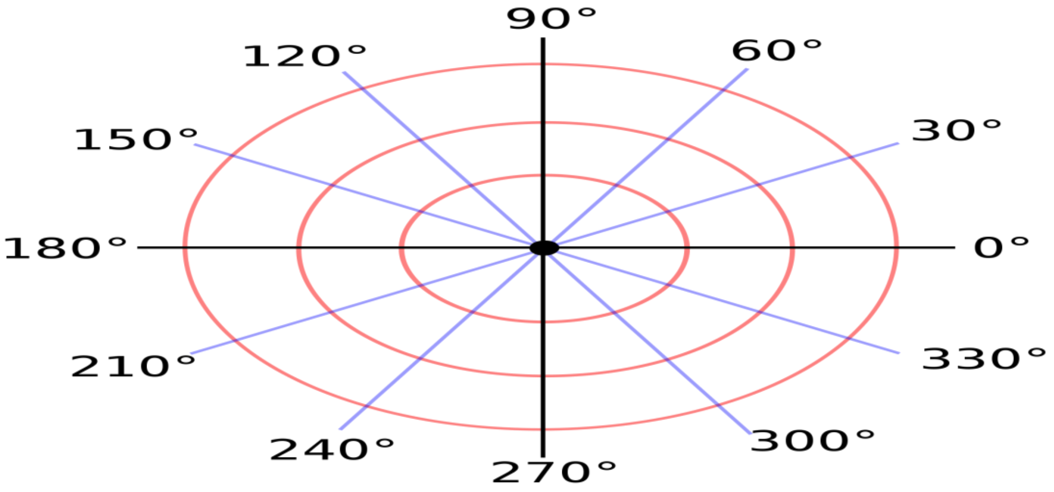
- Expressing pts on a **circular grid**
- Positions indicated by ordered pair
  - 'r' (distance from origin)
  - angle (from polar axis)

Pts can have infinite polar coordinates

Cartesian coordinates

Polar coordinates



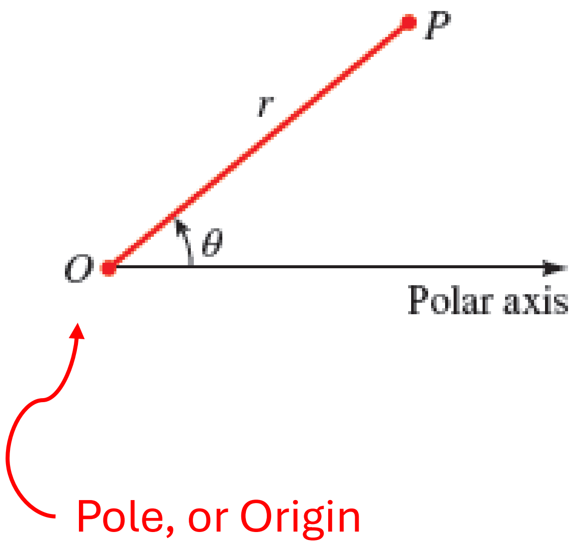


Rectangular grid

Grid consisting of **circles centered at the pole** and **rays emanating from the pole**

Definition of polar coordinates

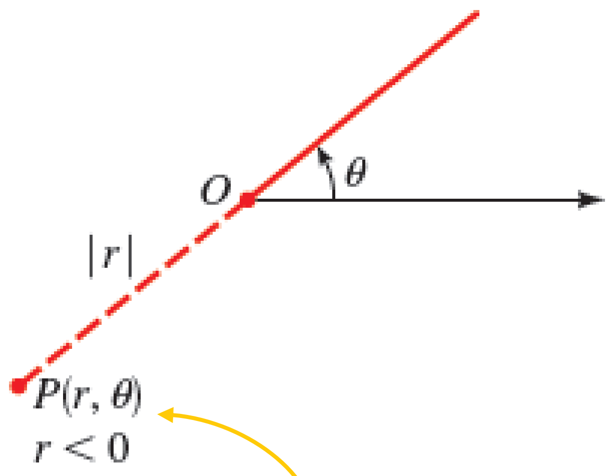
The **polar coordinate system** uses **distances** and **directions** to specify the location of a point in the plane.



Polar coordinates of a point consist of an ordered pair, **(r,θ)**, where

- r is the distance from the point to the origin, and
- θ is the angle measured in standard position

Conventions for negative angles and radii

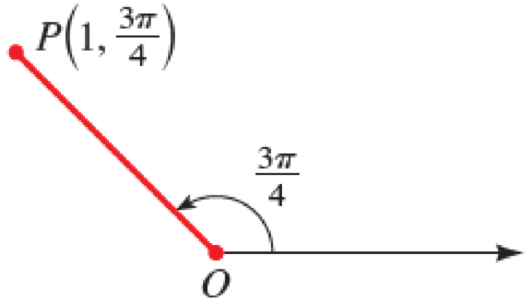


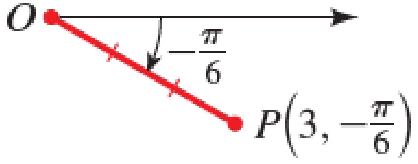
θ is

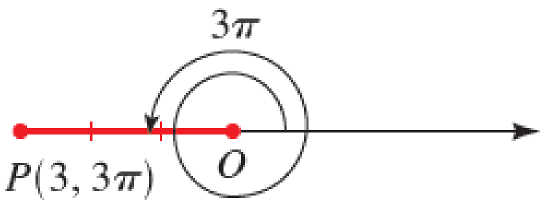
- **positive** if measured in a **counterclockwise direction** from the polar axis or
- **negative** if measured in a **clockwise direction**

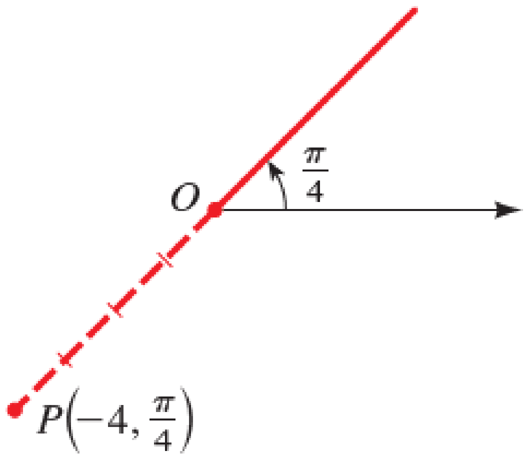
For **negative r values**
Move in the **opposite direction**.

Polar coordinates: Examples









$(1, \frac{3\pi}{4})$ $(3, \frac{-\pi}{6})$ $(3, 3\pi)$ $(-4, \frac{\pi}{4})$

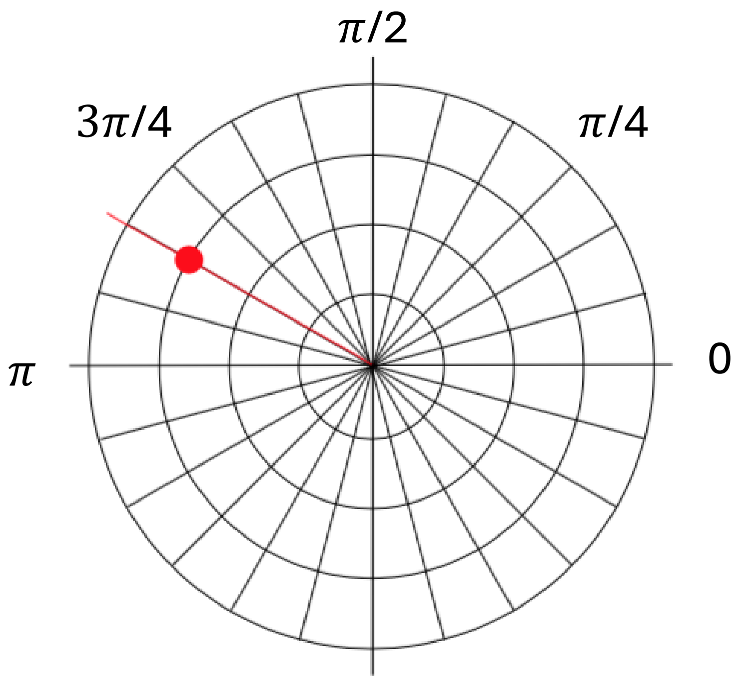
Plotting points in polar coordinates

Plot the polar point $(3, \dfrac{5\pi}{6})$.

Solution

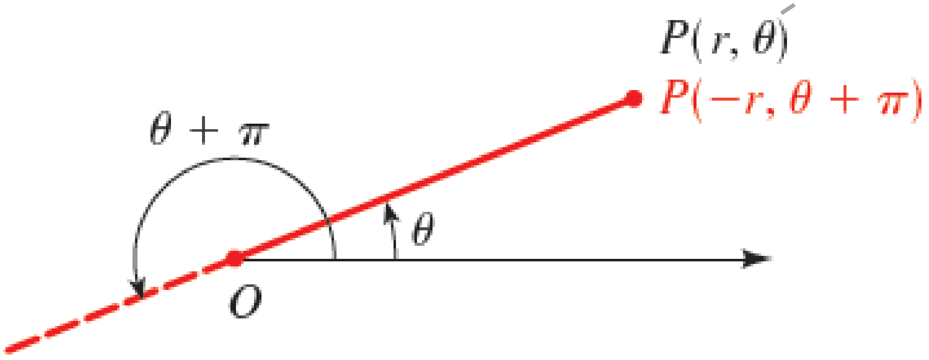
This point will be a distance of 3 from the origin, at an angle of $\frac{5\pi}{6}$.

Plotting this



EXAMPLE

Different polar coordinates for same point



Any point $P(r, \theta)$ can be represented by

$$\boldsymbol{P(r, \theta + 2n\pi)}$$

And

$$\boldsymbol{P(-r, \theta + (2n+1)\pi)}$$

Brings back to the same point!

Plotting points in polar coordinates

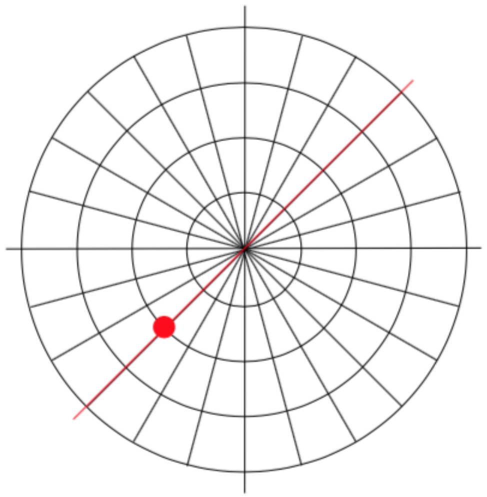
Plot the polar point $(-2, \frac{\pi}{4})$.

Solution

To start, we rotate to an angle of $\frac{\pi}{4}$.

Moving this direction, into the first quadrant, would be **positive *r* values**.

For **negative *r* values**, we **move in the opposite direction**, into the third quadrant. Plotting this



The resulting point is the same as the polar point $(2, \frac{5\pi}{4})$.

In fact, any Cartesian point can be represented by an infinite number of different polar coordinates by adding or subtracting full rotations to these points.

For example, same point could also be represented as $(2, \frac{13\pi}{4})$

EXAMPLE

Plotting points in polar coordinates

Plot each of the following points on the polar plane.

a. $(2, \frac{\pi}{4})$

b. $(-3, \frac{2\pi}{3})$

c. $(4, \frac{5\pi}{4})$

ACTIVITY 1.1

1.1 Plotting points in polar coordinates

SOLUTION

Different Polar Coordinates for the Same Point

Plot the point that has the given polar coordinates. Then give two other polar coordinate representations of the point, one with $r < 0$ and the other with $r > 0$.

(a) $(2, 3\pi/4)$

(b) $(-2, -\pi/3)$

(c) $(3, 1)$

ACTIVITY 1.2

1.2a Different Polar Coordinates for the Same Point

SOLUTION

1.2b Different Polar Coordinates for the Same Point

SOLUTION

1.2c Different Polar Coordinates for the Same Point

SOLUTION

SSILP Mathematics Course survey

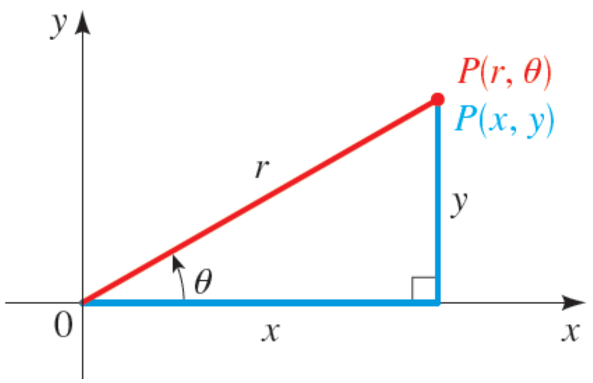
The SSILP Mathematics Survey is now open until XXXX, you should have received an **email** on this.

Please **follow the link provided in the email** to complete the survey.

Your feedback is very important for us to improve the course.

SOLUTION

Relationship between polar and rectangular coordinates



$$\cos\theta = \frac{x}{r} \quad \Rightarrow \quad x = r\cos\theta$$

$$\sin\theta = \frac{y}{r} \quad \Rightarrow \quad y = r\sin\theta$$

RELATIONSHIP BETWEEN POLAR AND RECTANGULAR COORDINATES

1. To change from polar to rectangular coordinates, use the formulas

$$x = r\cos\theta \qquad \text{and} \qquad y = r\sin\theta$$

2. To change from rectangular to polar coordinates, use the formulas

$$r^2 = x^2 + y^2 \qquad \text{and} \qquad \tan\theta = \frac{y}{x} \quad (x \neq 0)$$

Converting Polar Coordinates to Rectangular Coordinates

Find rectangular coordinates for the point that has polar coordinates $(4, 2\pi/3)$

SOLUTION Since $r = 4$ and $\theta = 2\pi/3$, we have

$$x = r\cos\theta = 4\cos\frac{2\pi}{3} = 4\cdot\left(-\frac{1}{2}\right) = -2$$

$$y = r\sin\theta = 4\sin\frac{2\pi}{3} = 4\cdot\frac{\sqrt{3}}{2} = 2\sqrt{3}$$

Thus the point has rectangular coordinates $(-2, 2\sqrt{3})$.

EXAMPLE

Converting Rectangular Coordinates to Polar Coordinates

Find polar coordinates for the point that has rectangular coordinates $(2, -2)$.

SOLUTION Using $x = 2$, $y = -2$, we get

$$r^2 = x^2 + y^2 = 2^2 + (-2)^2 = 8$$

so $r = 2\sqrt{2}$ or $-2\sqrt{2}$. Also

$$\tan\theta = \frac{y}{x} = \frac{-2}{2} = -1$$

so $\theta = 3\pi/4$ or $-\pi/4$. Since the point $(2, -2)$ lies in Quadrant IV (see Figure 7), we can represent it in polar coordinates as $(2\sqrt{2}, -\pi/4)$ or $(-2\sqrt{2}, 3\pi/4)$.

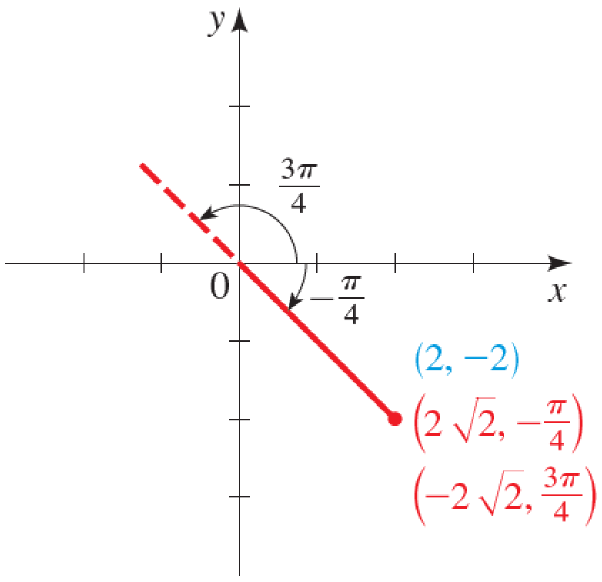


FIGURE 7

Note that the equations relating polar and rectangular coordinates **do not uniquely determine *r* or *θ*.**

When using these equations to find the polar coordinates of a point, be careful that the values chosen for ***r* or *θ*** give a point in **the correct quadrant**

EXAMPLE

Polar coordinates to rectangular coordinates

(a) $(6, 2\pi/3)$

(b) $(-1, 5\pi/2)$

ACTIVITY 1.3

1.3 Polar coordinates to rectangular coordinates

SOLUTION

Rectangular coordinates to polar coordinates

(a) $(-\sqrt{6}, -\sqrt{2})$

(b) $(0, -\sqrt{3})$

ACTIVITY 1.4

1.4 Rectangular coordinates to polar coordinates

SOLUTION

Polar equations

Converting an Equation from Rectangular to Polar Coordinates

Express the equation $x^2 = 4y$ in polar coordinates.

SOLUTION We use the formulas $x = r\cos\theta$ and $y = r\sin\theta$.

$$
\begin{aligned}
x^2 &= 4y && \text{Rectangular equation} \\
(r\cos\theta)^2 &= 4(r\sin\theta) && \text{Substitute } x = r\cos\theta,\ y = r\sin\theta \\
r^2\cos^2\theta &= 4r\sin\theta && \text{Expand} \\
r &= 4\frac{\sin\theta}{\cos^2\theta} && \text{Divide by } r\cos^2\theta \\
r &= 4\sec\theta\tan\theta && \text{Simplify}
\end{aligned}
$$

EXAMPLE

Converting Equations from Polar to Rectangular Coordinates

Express the polar equation in rectangular coordinates. If possible, determine the graph of the equation from its rectangular form.

(a) $r = 5 \sec \theta$

(a) Since $\sec \theta = 1/\cos \theta$, we multiply both sides by $\cos \theta$.

$$r = 5 \sec \theta \qquad \text{Polar equation}$$

$$r \cos \theta = 5 \qquad \text{Multiply by } \cos \theta$$

$$x = 5 \qquad \text{Substitute } x = r \cos \theta$$

The graph of $x = 5$ is the vertical line in Figure 8.

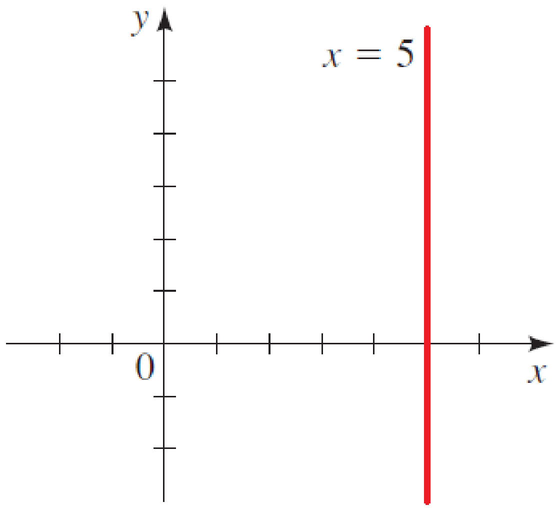


FIGURE 8

EXAMPLE

Converting Equations from Polar to Rectangular Coordinates

Express the polar equation in rectangular coordinates. If possible, determine the graph of the equation from its rectangular form.

(b) $r = 2 \sin \theta$

(b) We multiply both sides of the equation by r, because then we can use the formulas $r^2 = x^2 + y^2$ and $r \sin \theta = y$.

$$
\begin{aligned}
r &= 2 \sin \theta && \text{Polar equation} \\
r^2 &= 2r \sin \theta && \text{Multiply by } r \\
x^2 + y^2 &= 2y && r^2 = x^2 + y^2 \text{ and } r \sin \theta = y \\
x^2 + y^2 - 2y &= 0 && \text{Subtract } 2y \\
x^2 + (y - 1)^2 &= 1 && \text{Complete the square in } y
\end{aligned}
$$

This is the equation of a circle of radius 1 centered at the point $(0, 1)$. It is graphed in Figure 9.

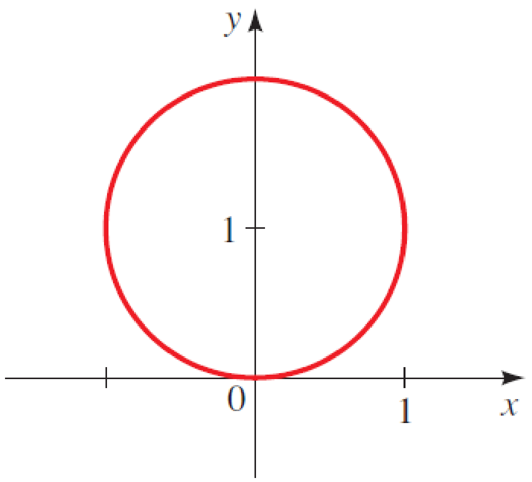


FIGURE 9

EXAMPLE

Rectangular equations to polar equations

(a) $y = 5$

(b) $x^2 - y^2 = 1$

ACTIVITY 1.5

1.5 Rectangular equations to polar equations

SOLUTION

Polar equations to rectangular equations

Convert from polar form to rectangular form

(a) $r = 4\cos\theta$

(b) $r = 8\sin\theta$

(c) $r^2 = 4\sin(2\theta)$

ACTIVITY 1.6

1.6 Polar equations to rectangular equations

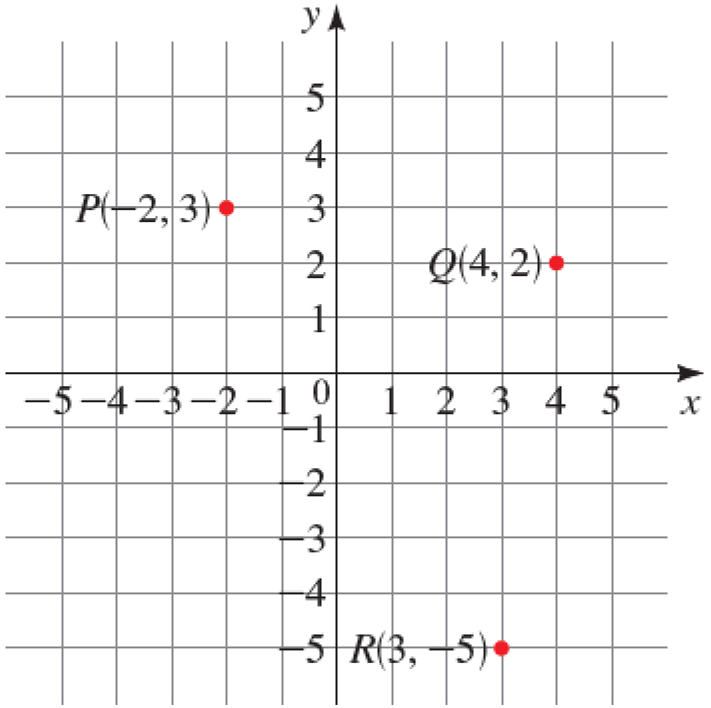
SOLUTION

1.6 Polar equations to rectangular equations

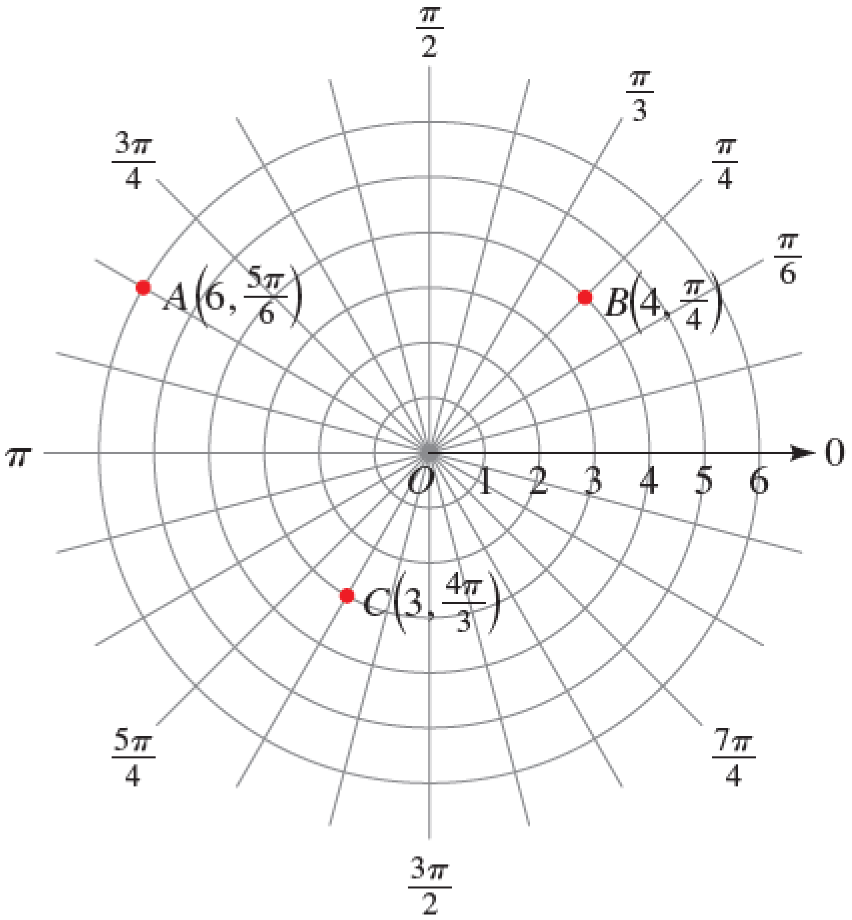
SOLUTION

2. Graphs of polar equations

Let’s have a closer look at the grids



(a) Grid for rectangular coordinates



(b) Grid for polar coordinates

Sketching the graph of a polar equation

Sketch a graph of the equation $r = 3$, and express the equation in rectangular coordinates.

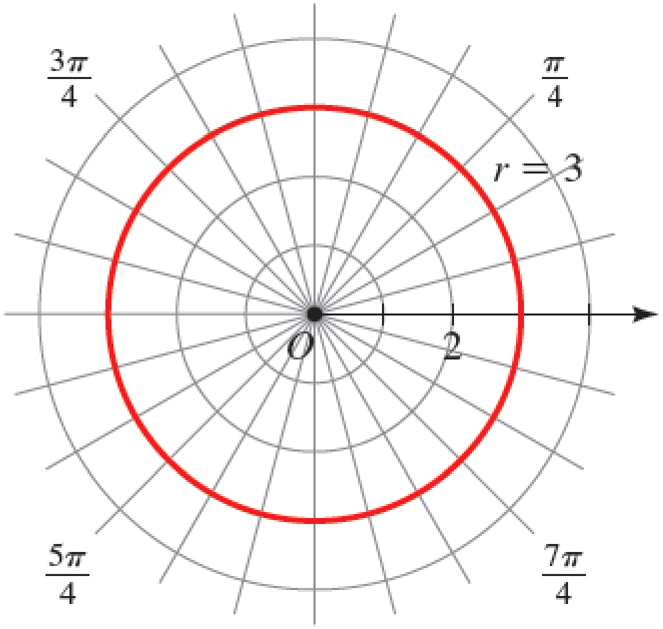
SOLUTION The graph consists of all points whose r-coordinate is 3, that is, all points that are 3 units away from the origin. So the graph is a circle of radius 3 centered at the origin, as shown in Figure 2.

Squaring both sides of the equation, we get

$$r^2 = 3^2 \qquad \text{Square both sides}$$

$$x^2 + y^2 = 9 \qquad \text{Substitute } r^2 = x^2 + y^2$$

So the equivalent equation in rectangular coordinates is $x^2 + y^2 = 9$.



EXAMPLE

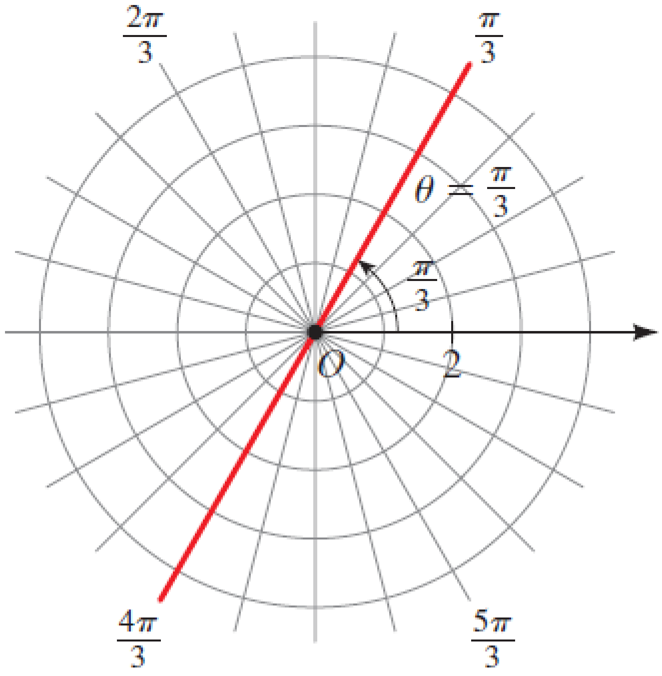
Sketching the graph of a polar equation

Sketch a graph of the equation $\theta = \pi/3$, and express the equation in rectangular coordinates.

SOLUTION The graph consists of all points whose θ-coordinate is $\pi/3$. This is the straight line that passes through the origin and makes an angle of $\pi/3$ with the polar axis (see Figure 3). Note that the points $(r, \pi/3)$ on the line with $r > 0$ lie in Quadrant I, whereas those with $r < 0$ lie in Quadrant III. If the point (x, y) lies on this line, then

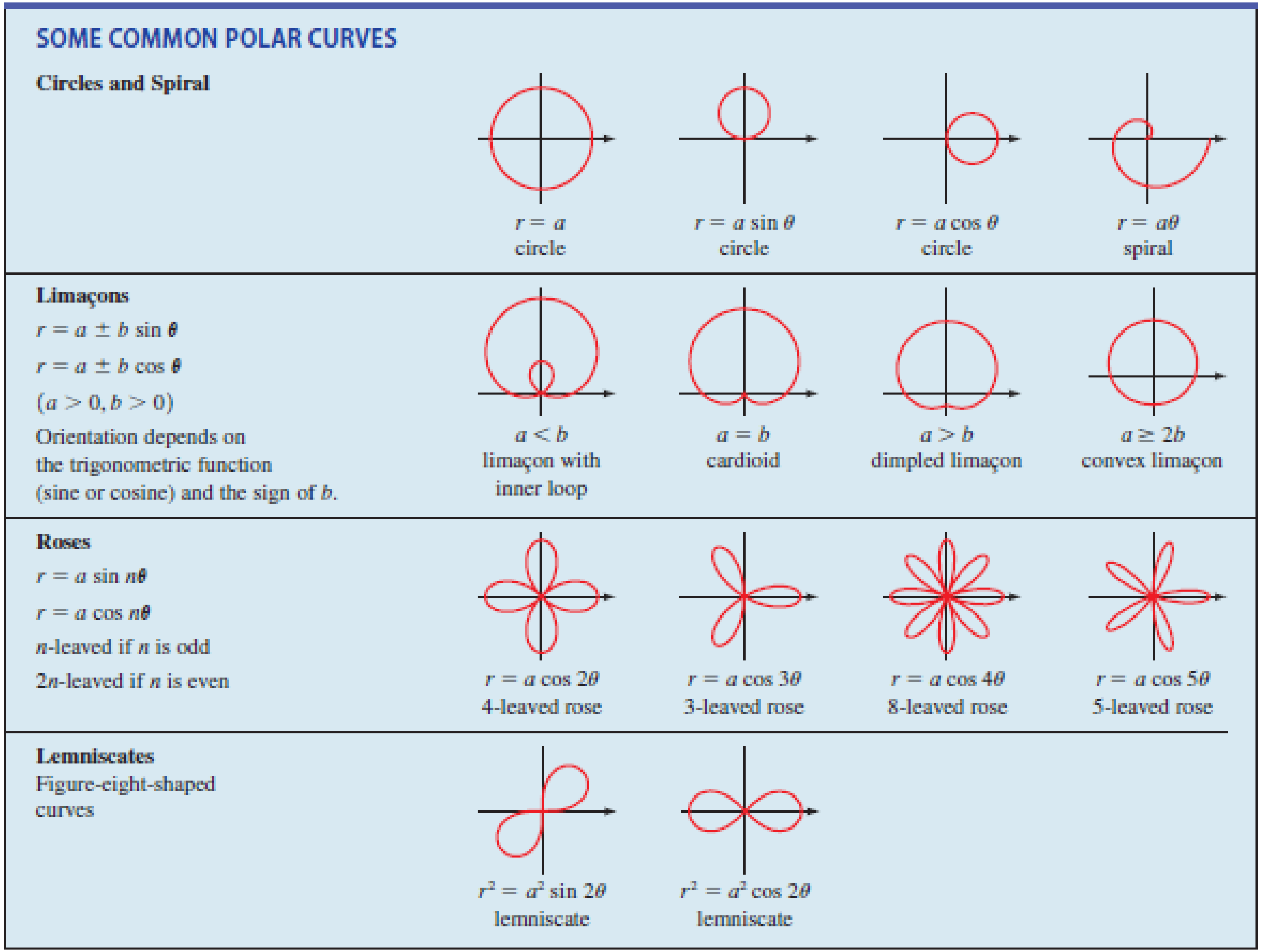
$$\frac{y}{x} = \tan\theta = \tan\frac{\pi}{3} = \sqrt{3}$$

Thus the rectangular equation of this line is $y = \sqrt{3}x$.



EXAMPLE

Some common polar curves



Polar to rectangular

Sketch a graph of the polar equation, and express the equation in rectangular coordinates

(a) $r = 2$

(b) $\theta = -\pi/2$

(c) $\theta = 5\pi/6$

(d) $r = \cos\theta$

ACTIVITY 2.1

Polar to rectangular

SOLUTION